KCE PATHWAYS · GRADE 3 · UNIT 4: PARTS AND PIECES · WEEK 16 · MATH

Slice Night at Russo's Pizza Kitchen — Teacher Lesson Plan & Answer Key

Learning objectives: Students check that a whole is split into equal parts before naming them (halves, thirds, fourths, sixths, eighths), name one equal part with a unit fraction, build fractions like 5/8 from unit fractions, explain what the numerator and denominator tell, and write fractions of a whole and of a group, including two-step and find-the-mistake problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Slice Night at Russo's Pizza Kitchen in Brooklyn, New York, with Mr. Russo, Owen, Hana and Darius.
2. Warm-Up, Part 1: Division and Equal Groups (8 pts, Unit 3 skill): a typed division fact and a fact-family reasoning item. If a student misses both, model "6 times what is 24?" before moving on.
3. Warm-Up, Part 2 (4 pts): a two-step division problem. Ask the student to say step 1 out loud before typing.
4. Picture It: Slice Night at Russo's Pizza Kitchen. Walk through Mr. Russo's job list (cut pizzas into equal slices, fill topping trays, pack garlic knots). Ask: how do you know two slices are the same size?
5. Word Wrangler: 6 vocab cards (whole, equal parts, fraction, unit fraction, numerator, denominator), each with a sentence from the story. Have the student say each meaning back.
6. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W16.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
7. Equal Parts of a Whole, Part 1 (8 pts): picking the shape with equal parts, naming parts (halves, thirds, fourths), and find-the-mistake. Tip: before naming any part, ask "Are all the parts the same size?"
8. Equal Parts of a Whole, Part 2 (4 pts): a two-step cutting problem or a strip of dough split into thirds or fourths. Tip: sketch the strip and mark the cuts.
9. Unit Fractions, Part 1 (8 pts): naming one slice with a unit fraction (typed with a slash, like 1/8) and putting unit fractions together (five 1/8 pieces make 5/8). Tip: count the pieces aloud, "one eighth, two eighths...".
10. Unit Fractions, Part 2 (4 pts): which-explanation-is-correct and find-the-mistake about what makes a unit fraction (the numerator is 1).
11. Numerator and Denominator, Part 1 (8 pts): writing the shaded fraction and two-step "what fraction does NOT have..." items. Tip: count ALL the equal parts first for the denominator, then the parts you want for the numerator.
12. Numerator and Denominator, Part 2 (4 pts): reasoning about the denominator and why 4/4 is one whole.
13. Fractions of a Group, Part 1 (8 pts): fractions of olives, tomatoes, boxes and jars, including two-step items. Tip: the whole is the whole group, so the denominator is how many items in all.

14. Fractions of a Group, Part 2 (4 pts): find-the-mistake items where the student compared a part to the other part instead of to the whole group.
15. Bonus Practice, Part 1: The Pizza Counter (XP only, not scored).
16. Bonus Practice, Part 2: Slices and Olives (XP only, not scored).
17. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches “3 of 8 equal parts” to 3/8; Set 2 matches fraction vocabulary to meanings; Set 3 matches groups and part names.
18. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters, names, end mark, list commas); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
19. Speed Arena, Part 1 (XP only, not scored).
20. Speed Arena, Part 2 and Keyboard Corral (XP only, not scored).
21. Boss Battle, Part 1: The Big Slice Night (10 pts, 5 each): a two-step typed fraction problem and a which-explanation-is-correct item.
22. Boss Battle, Part 2: Closing the Kitchen (10 pts, 5 each): a two-step equal-groups problem and a find-the-mistake item (for example, 5/30 instead of 5/6).
23. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W16.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
24. Results (Slice Night Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Slice Night Hero!” with the congratulations message “Slice Night hero! You named every part of the whole!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every pizza maker learns one slice at a time.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: DIVISION AND EQUAL GROUPS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
`w1_s0`	Set 1: Mr. Russo has 24 paper plates. He puts them into 6 equal stacks. How many plates are in each stack?	**4** (also accepted: four, 4 plates, 4 plate, 4 paper plates)	Division splits 24 into 6 equal stacks: 24 ÷ 6 = 4, because 6 × 4 = 24.
`w2_s0`	Set 1: Which explanation is correct? How does 8 × 7 = 56 help you solve 56 ÷ 7? [They are in the same fact family, so 56 ÷ 7 = 8. / You add the numbers: 56 + 7 = 63. / It does not help. Multiplication and division are not related. / It shows that 56 ÷ 7 = 7.]	**They are in the same fact family, so 56 ÷ 7 = 8.**	Multiplication and division facts with the same three numbers belong to one fact family. 8 × 7 = 56, so 56 ÷ 7 = 8.
`w3_s0`	Set 1: Two steps! Owen has 30 garlic knots. He packs them into boxes that hold 6 knots each. Then he gives 2 full boxes to a neighbor. How many boxes does he have left?	**3** (also accepted: three, 3 boxes, 3 box)	Step 1: 30 ÷ 6 = 5 boxes. Step 2: 5 − 2 = 3 boxes left.
`w1_s1`	Set 2: Hana rolls a rope of dough 42 inches long and cuts it into 7 equal pieces. How long is each piece, in inches?	**6** (also accepted: six, 6 inches, 6 inch, 6 in, 6 in.)	42 ÷ 7 = 6, because 7 × 6 = 42. Each piece is 6 inches long.

ID	Question	Answer	Why (explain to student)
w2_s1	Set 2: Find the mistake. Darius says 36 ÷ 4 = 8, because 4 × 8 = 36. What went wrong? [4 × 8 is 32, not 36. Since 4 × 9 = 36, the answer is 9. / Nothing is wrong. 36 ÷ 4 = 8. / He should subtract: 36 − 4 = 32. / He should add: 36 + 4 = 40.]	**4 × 8 is 32, not 36. Since 4 × 9 = 36, the answer is 9.**	4 × 8 = 32, not 36. The fact 4 × 9 = 36 shows that 36 ÷ 4 = 9.
w3_s1	Set 2: Two steps! Mr. Russo bakes 4 pizzas and cuts each pizza into 6 equal slices. He shares all the slices equally among 8 people. How many slices does each person get?	**3** (also accepted: three, 3 slices, 3 slice)	Step 1: 4 pizzas × 6 slices = 24 slices. Step 2: 24 ÷ 8 = 3 slices each.
w1_s2	Set 3: Mr. Russo packs 54 jars of tomato sauce into 9 boxes, with the same number in each box. How many jars go in each box?	**6** (also accepted: six, 6 jars, 6 jar)	54 ÷ 9 = 6, because 9 × 6 = 54.
w2_s2	Set 3: Which fact belongs in the same fact family as 7 × 9 = 63? [63 ÷ 9 = 7 / 63 − 9 = 54 / 7 + 9 = 16 / 9 ÷ 7 = 63]	**63 ÷ 9 = 7**	A fact family uses the same three numbers with multiplication and division. 63 ÷ 9 = 7 uses 7, 9, and 63.
w3_s2	Set 3: Two steps! Darius unpacks 48 tomatoes. He puts 8 tomatoes in each crate. Then he uses 2 crates to make sauce. How many crates are left?	**4** (also accepted: four, 4 crates, 4 crate)	Step 1: 48 ÷ 8 = 6 crates. Step 2: 6 − 2 = 4 crates left.

EQUAL PARTS OF A WHOLE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p1a_s0	Set 1: Mr. Russo wants a shape cut into equal parts. Which shape is it? Type the letter.	**C** (also accepted: c, shape C, shape c)	Shape C is cut into 3 parts that are all the same size. The parts in A and B are different sizes, and D has a big part and a small part.
p1b_s0	Set 1: Find the mistake. Darius cut a pizza into 4 pieces: 2 big pieces and 2 small pieces. He says each piece is one fourth of the pizza. What went wrong? [The pieces are not the same size, so they are not fourths. / Nothing is wrong. Any 4 pieces are fourths. / He should call them halves, because there are 2 big pieces. / Each piece is one eighth of the pizza.]	**The pieces are not the same size, so they are not fourths.**	Fourths must be 4 parts that are all the same size. 2 big pieces and 2 small pieces are not equal, so they are not fourths.
p1c_s0	Set 1: Two steps! Owen cuts a pan of Sicilian pizza into 3 equal rows. Then he cuts each row into 2 equal pieces. What do we call each piece? Type the name of the parts, like halves or thirds.	**sixths** (also accepted: sixth, 6ths, one sixth, 1/6, a sixth, 6 sixths, six sixths)	Step 1: 3 rows × 2 pieces = 6 equal parts. Step 2: 6 equal parts are called sixths.
p1a_s1	Set 2: This long flatbread is cut into equal parts. What do we call each part? Type the name of the parts, like halves or thirds.	**eighths** (also accepted: eighth, 8ths, one eighth, 1/8, an eighth, 8 eighths, eight eighths)	Count the equal parts: there are 8. Eight equal parts are called eighths.

ID	Question	Answer	Why (explain to student)
p1b_s1	Set 2: Which explanation is correct? A square is cut from corner to corner into 4 triangles that are all the same size. Is each triangle one fourth of the square? [Yes. The 4 parts are the same size, so each one is a fourth. / No. Fourths must always be small squares. / No. Triangles can only be halves. / Yes, because any 4 pieces are always fourths.]	**Yes. The 4 parts are the same size, so each one is a fourth.**	Equal parts must be the same size, but they can be any shape. The 4 triangles are the same size, so each one is a fourth.
p1c_s1	Set 2: Hana has a strip of dough 12 inches long. She cuts it into thirds. How long is each part, in inches?	**4** (also accepted: four, 4 inches, 4 inch, 4 in, 4 in.)	Thirds are 3 equal parts. 12 ÷ 3 = 4, so each part is 4 inches long.
p1a_s2	Set 3: Which shape is cut into thirds? Type the letter.	**A** (also accepted: a, shape A, shape a)	Thirds are 3 equal parts. Only shape A has 3 parts that are all the same size. B has 3 parts of different sizes, C has 4 parts, and D has 2 parts.
p1b_s2	Set 3: Find the mistake. Owen folded a square paper napkin in half, and then in half again. He says the parts are thirds, because he folded it 2 times. What went wrong? [Folding in half two times makes 4 equal parts, so they are fourths. / Nothing is wrong. Two folds make thirds. / The parts are halves, because he folded it in half. / The parts are eighths.]	**Folding in half two times makes 4 equal parts, so they are fourths.**	Folding in half makes 2 parts. Folding in half again cuts each of those in half, so there are 4 equal parts: fourths.
p1c_s2	Set 3: Mr. Russo cuts a rope of dough 16 inches long into fourths. How long is each part, in inches?	**4** (also accepted: four, 4 inches, 4 inch, 4 in, 4 in.)	Fourths are 4 equal parts. 16 ÷ 4 = 4, so each part is 4 inches long.

UNIT FRACTIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
u1a_s0	Set 1: Every slice of this pizza is the same size. What unit fraction names ONE slice? Type it with a slash, like 1/2.	**1/6** (also accepted: 1 / 6, one sixth, one-sixth, 1 sixth, 1 out of 6, a sixth)	The pizza is cut into 6 equal slices, so one slice is 1 of 6 equal parts: 1/6.
u1b_s0	Set 1: Hana puts pepperoni on 5 parts of a pizza. Each part is 1/8 of the pizza. What fraction of the pizza did she cover? Type it like 3/4.	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths, 5 out of 8)	Each piece is 1/8. Count them: one eighth, two eighths, three eighths, four eighths, five eighths. That is 5/8.
u1c_s0	Set 1: Which explanation is correct? Why is 1/4 a unit fraction, but 3/4 is not? [1/4 names just one equal part. 3/4 names three equal parts. / 1/4 has a smaller bottom number. / 3/4 is more than one whole. / Unit fractions always have a 4 on the bottom.]	**1/4 names just one equal part. 3/4 names three equal parts.**	A unit fraction names just one equal part, so its top number is 1. 3/4 names three parts.
u1a_s1	Set 2: This flatbread is cut into equal parts. What unit fraction names ONE part? Type it with a slash, like 1/2.	**1/3** (also accepted: 1 / 3, one third, one-third, 1 third, 1 out of 3, a third)	The flatbread is cut into 3 equal parts, so one part is 1/3.
u1b_s1	Set 2: A pizza at the counter is cut into 6 equal slices, so each slice is 1/6 of the pizza. Darius sold 4/6 of the pizza. How many slices did he sell?	**4** (also accepted: four, 4 slices, 4 slice)	4/6 means 4 slices that are each 1/6 of the pizza. Darius sold 4 slices.

ID	Question	Answer	Why (explain to student)
u1c_s1	Set 2: Find the mistake. Owen says 1/3 and 2/3 are both unit fractions, because they both have a 3 on the bottom. What went wrong? [A unit fraction has a 1 on top. 2/3 names two parts, so it is not a unit fraction. / Nothing is wrong. Any fraction with a 3 on the bottom is a unit fraction. / Neither one is a unit fraction. / Only 2/3 is a unit fraction, because 2 is bigger.]	**A unit fraction has a 1 on top. 2/3 names two parts, so it is not a unit fraction.**	A unit fraction has a 1 on top. 2/3 names two equal parts, so it is not a unit fraction.
u1a_s2	Set 3: This square pizza is cut into equal parts. What unit fraction names ONE shaded part? Type it with a slash, like 1/2.	**1/4** (also accepted: 1 / 4, one fourth, one-fourth, 1 fourth, 1 out of 4, a fourth, one quarter, one-quarter, 1 quarter, a quarter)	The pizza is cut into 4 equal parts, so one part is 1/4.
u1b_s2	Set 3: Two steps! A square pizza is cut into 8 equal parts, and each part is 1/8 of the pizza. Hana has topped 3 parts with peppers. How many more 1/8 parts does she need to top to finish the whole pizza?	**5** (also accepted: five, 5 parts, 5 part, 5 pieces, 5 piece)	Step 1: the whole pizza is 8 parts of 1/8. Step 2: 8 − 3 = 5 more parts.
u1c_s2	Set 3: Which explanation is correct? Mr. Russo says 7/8 is the same as seven 1/8 pieces put together. [He is right. 7/8 means 7 parts that are each 1/8 of the whole. / He is wrong. 7/8 means 7 wholes and 8 parts. / He is wrong. 7/8 means 7 + 8 = 15 parts. / He is wrong. You can never put unit fractions together.]	**He is right. 7/8 means 7 parts that are each 1/8 of the whole.**	7/8 means 7 parts that are each 1/8 of the whole, so seven 1/8 pieces make 7/8.

NUMERATOR AND DENOMINATOR (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1a_s0	Set 1: What fraction of this flatbread is shaded? Type it with a slash, like 3/4.	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths, 5 out of 8)	There are 8 equal parts in the whole flatbread (the denominator) and 5 are shaded (the numerator): 5/8.
c1b_s0	Set 1: Two steps! A square pizza has 6 equal parts. 2 parts have pepperoni. What fraction of the pizza does NOT have pepperoni? Type it like 3/4.	**4/6** (also accepted: 4 / 6, four sixths, four-sixths, 4 sixths, 4 out of 6)	Step 1: 6 − 2 = 4 parts have no pepperoni. Step 2: 4 of 6 equal parts is 4/6.
c1c_s0	Set 1: Find the mistake. A circle is cut into 4 equal parts, and 3 parts are shaded. Darius says the shaded part is 3/1, because 3 parts are shaded and 1 is not. What went wrong? [The denominator must count all 4 equal parts. The shaded part is 3/4. / Nothing is wrong. The shaded part is 3/1. / The shaded part is 1/3. / The shaded part is 4/3.]	**The denominator must count all 4 equal parts. The shaded part is 3/4.**	The denominator counts ALL the equal parts, shaded and not shaded. 3 of 4 equal parts are shaded, so the fraction is 3/4.
c1a_s1	Set 2: What fraction of this pizza is shaded? Type it with a slash, like 3/4.	**4/6** (also accepted: 4 / 6, four sixths, four-sixths, 4 sixths, 4 out of 6)	The pizza has 6 equal slices (the denominator) and 4 are shaded (the numerator): 4/6.
c1b_s1	Set 2: Two steps! A square pizza has 8 equal parts. 3 parts have mushrooms, 2 parts have peppers, and the rest are plain cheese. What fraction of the pizza is plain cheese? Type it like 3/4.	**3/8** (also accepted: 3 / 8, three eighths, three-eighths, 3 eighths, 3 out of 8)	Step 1: 3 + 2 = 5 parts have mushrooms or peppers. Step 2: 8 − 5 = 3 plain cheese parts, so plain cheese is 3/8.

ID	Question	Answer	Why (explain to student)
c1c_s1	Set 2: Which explanation is correct? What does the denominator of 5/6 tell you? [The whole is cut into 6 equal parts. / 5 parts are being counted. / There are 6 whole pizzas. / You add 5 and 6 to get 11 parts.]	**The whole is cut into 6 equal parts.**	The denominator is the bottom number. In 5/6, it tells you the whole is cut into 6 equal parts.
c1a_s2	Set 3: What fraction of this tray pizza is shaded? Type it with a slash, like 3/4.	**2/3** (also accepted: 2 / 3, two thirds, two-thirds, 2 thirds, 2 out of 3)	There are 3 equal parts (the denominator) and 2 are shaded (the numerator): 2/3.
c1b_s2	Set 3: Two steps! A tray pizza has 6 equal parts. Hana tops 1/6 of it with spinach. Then she tops 3 more parts with spinach. What fraction of the tray pizza has spinach now? Type it like 3/4.	**4/6** (also accepted: 4 / 6, four sixths, four-sixths, 4 sixths, 4 out of 6)	Step 1: 1 part + 3 more parts = 4 spinach parts. Step 2: 4 of 6 equal parts is 4/6.
c1c_s2	Set 3: Find the mistake. Hana says 4/4 of a square pizza is less than the whole pizza, because 4 is a small number. What went wrong? [4/4 means all 4 of the 4 equal parts, so 4/4 is one whole pizza. / Nothing is wrong. 4/4 is less than one whole. / 4/4 is the same as 4 whole pizzas. / 4/4 is the same as 1/4.]	**4/4 means all 4 of the 4 equal parts, so 4/4 is one whole pizza.**	4/4 means all 4 of the 4 equal parts, so 4/4 is the same as one whole.

FRACTIONS OF A GROUP (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1a_s0	Set 1: What fraction of these tomatoes are green? Type it with a slash, like 3/4.	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths, 5 out of 8)	There are 8 tomatoes in the whole group and 5 are green, so 5/8 are green.
d1b_s0	Set 1: Two steps! Hana stacks 6 pizza boxes. 1 is red, 2 are blue, and the rest are white. What fraction of the boxes are white? Type it like 3/4.	**3/6** (also accepted: 3 / 6, three sixths, three-sixths, 3 sixths, 3 out of 6)	Step 1: 1 + 2 = 3 boxes are red or blue. Step 2: 6 − 3 = 3 white boxes, so 3/6 are white.
d1c_s0	Set 1: Find the mistake. Darius has 4 peppers: 1 orange and 3 green. He says 1/3 of the peppers are orange. What went wrong? [He compared orange to green. The whole group is 4 peppers, so 1/4 are orange. / Nothing is wrong. 1/3 of the peppers are orange. / 3/4 of the peppers are orange. / 1/1 of the peppers are orange.]	**He compared orange to green. The whole group is 4 peppers, so 1/4 are orange.**	For a set, the denominator is the number of things in the whole group: 4 peppers. 1 of 4 is orange, so 1/4.
d1a_s1	Set 2: What fraction of the olives on this tray are black? Type it with a slash, like 3/4.	**1/6** (also accepted: 1 / 6, one sixth, one-sixth, 1 sixth, 1 out of 6, a sixth)	There are 6 olives on the whole tray and 1 is black, so 1/6 are black.
d1b_s1	Set 2: Mr. Russo is baking 8 pizzas. 3/8 of them are out of the oven. How many pizzas are still in the oven?	**5** (also accepted: five, 5 pizzas, 5 pizza)	3/8 out of the oven means 3 of the 8 pizzas are out. 8 − 3 = 5 pizzas are still in the oven.
d1c_s1	Set 2: Which explanation is correct? Owen has 3 tomatoes: 1 big tomato and 2 small tomatoes. The big tomato is red. Is 1/3 of the group of tomatoes red? [Yes. In a group, each tomato counts as 1 of the 3, even if the sizes are different. / No. The big tomato is bigger, so it is 1/2 of the group. / No. The red tomato is 3/1 of the group. / No. You can only use fractions with pizzas and shapes.]	**Yes. In a group, each tomato counts as 1 of the 3, even if the sizes are different.**	In a set, each object counts as one part of the group, even if the objects are different sizes. 1 of the 3 tomatoes is red, so 1/3.

ID	Question	Answer	Why (explain to student)
d1a_s2	Set 3: The gold jars hold olive oil and the red jars hold tomato sauce. What fraction of the jars hold olive oil? Type it with a slash, like 3/4.	**4/6** (also accepted: 4 / 6, four sixths, four-sixths, 4 sixths, 4 out of 6)	There are 6 jars in the whole group and 4 hold olive oil, so 4/6 hold olive oil.
d1b_s2	Set 3: Two steps! A box holds 8 garlic knots. Darius eats 2 knots and gives 1 knot away. What fraction of the 8 garlic knots are still in the box? Type it like 3/4.	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths, 5 out of 8)	Step 1: 2 + 1 = 3 garlic knots are gone. Step 2: 8 − 3 = 5 knots left, so 5/8 of the garlic knots are still in the box.
d1c_s2	Set 3: Find the mistake. Hana has 6 slices on a tray: 2 cheese slices and 4 pepperoni slices. She says 2/4 of the slices are cheese. What went wrong? [She used only the pepperoni slices as the whole. All 6 slices are the whole, so 2/6 are cheese. / Nothing is wrong. 2/4 of the slices are cheese. / 4/2 of the slices are cheese. / 2/8 of the slices are cheese.]	**She used only the pepperoni slices as the whole. All 6 slices are the whole, so 2/6 are cheese.**	The whole group is all 6 slices, so the denominator is 6. 2 of 6 are cheese: 2/6.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 1 of 4 equal parts	**1/4**	2 points if matched on the first try.
match_s0_2	Set 1: Match 3 of 8 equal parts	**3/8**	2 points if matched on the first try.
match_s0_3	Set 1: Match 1 of 6 equal parts	**1/6**	2 points if matched on the first try.
match_s0_4	Set 1: Match 2 of 3 equal parts	**2/3**	2 points if matched on the first try.
match_s0_5	Set 1: Match 4 of 4 equal parts	**one whole**	2 points if matched on the first try.
match_s1_1	Set 2: Match numerator	**the top number: the parts we count**	2 points if matched on the first try.
match_s1_2	Set 2: Match denominator	**the bottom number: the equal parts in the whole**	2 points if matched on the first try.
match_s1_3	Set 2: Match unit fraction	**names just one equal part, like 1/3**	2 points if matched on the first try.
match_s1_4	Set 2: Match equal parts	**parts that are the same size**	2 points if matched on the first try.
match_s1_5	Set 2: Match whole	**one complete thing or group**	2 points if matched on the first try.
match_s2_1	Set 3: Match 2 green olives out of 6 olives	**2/6**	2 points if matched on the first try.
match_s2_2	Set 3: Match 5 red tomatoes out of 8 tomatoes	**5/8**	2 points if matched on the first try.
match_s2_3	Set 3: Match halves	**2 equal parts**	2 points if matched on the first try.
match_s2_4	Set 3: Match thirds	**3 equal parts**	2 points if matched on the first try.
match_s2_5	Set 3: Match eighths	**8 equal parts**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: owen sold slices at the counter	**Owen sold slices at the counter.**	Owen is a name and starts the sentence, so it gets a capital O. A telling sentence ends with a period.

ID	Question	Answer	Why (explain to student)
fs2_s0	Set 1: Fix the sentence: we baked cheese pepperoni and mushroom pizzas	**We baked cheese, pepperoni, and mushroom pizzas.** (also accepted: We baked cheese, pepperoni and mushroom pizzas.)	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how many olives are on the tray	**How many olives are on the tray?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: hana darius and owen cut the pizza	**Hana, Darius, and Owen cut the pizza.** (also accepted: Hana, Darius and Owen cut the pizza.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: the kitchen is in brooklyn	**The kitchen is in Brooklyn.**	Start with a capital T. Brooklyn is the name of a place, so it gets a capital B. End the telling sentence with a period.
fs2_s2	Set 3: Fix the sentence: we used red green and yellow peppers	**We used red, green, and yellow peppers.** (also accepted: We used red, green and yellow peppers.)	Start with a capital W, put commas between red, green, and yellow, and end with a period.

BOSS BATTLE: THE BIG SLICE NIGHT (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! Mr. Russo cuts a pizza into 8 equal slices. Owen eats 1/8 of the pizza and Hana eats 2/8. What fraction of the pizza is left? Type it like 3/4.	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths, 5 out of 8)	Step 1: 1/8 + 2/8 = 3 slices eaten. Step 2: 8 − 3 = 5 slices left, so 5/8 of the pizza is left.
bb2_s0	Set 1: Which explanation is correct? Owen cut a square into 4 parts that are NOT the same size and shaded 2 of them. Can he say 2/4 of the square is shaded? [No. Fourths must be equal parts, so 2/4 does not name the shaded part. / Yes. Any 2 parts out of 4 parts is 2/4. / Yes, because 2 is half of 4. / No, because the shaded part is 4/2.]	**No. Fourths must be equal parts, so 2/4 does not name the shaded part.**	Fractions name equal parts. The 4 parts are not the same size, so 2/4 does not name the shaded part.
bb1_s1	Set 2: Two steps! A tray pizza has 6 equal parts. 2 parts have pepperoni. Hana adds pepperoni to 1 more part. What fraction of the tray pizza does NOT have pepperoni now? Type it like 3/4.	**3/6** (also accepted: 3 / 6, three sixths, three-sixths, 3 sixths, 3 out of 6)	Step 1: 2 + 1 = 3 pepperoni parts. Step 2: 6 − 3 = 3 parts have no pepperoni, so 3/6 has no pepperoni.
bb2_s1	Set 2: Which explanation is correct? Mr. Russo says 6/6 of a pizza and 1 whole pizza are the same amount. [He is right. 6/6 means all 6 of the 6 equal slices, which is the whole pizza. / He is wrong. 6/6 is 6 whole pizzas. / He is wrong. 6/6 is less than one pizza. / He is wrong. 6/6 is the same as 1/6.]	**He is right. 6/6 means all 6 of the 6 equal slices, which is the whole pizza.**	6/6 means all 6 of the 6 equal slices, which is the whole pizza.

ID	Question	Answer	Why (explain to student)
bb1_s2	Set 3: Two steps! A pizza is cut into 8 equal slices. 2 slices are sold at lunch and 4 slices are sold at dinner. What fraction of the pizza is left? Type it like 3/4.	**2/8** (also accepted: 2 / 8, two eighths, two-eighths, 2 eighths, 2 out of 8)	Step 1: 2 + 4 = 6 slices sold. Step 2: 8 − 6 = 2 slices left, so 2/8 of the pizza is left.
bb2_s2	Set 3: Which explanation is correct? A tray has 6 olives: 3 green and 3 black. Darius says 3/6 of the olives are green. Hana says 3/3 of the olives are green. Who is right? [Darius. The whole group is all 6 olives, so 3/6 are green. / Hana. There are 3 green olives and 3 black olives, so it is 3/3. / Both of them are right. / Neither one. The answer is 6/3.]	**Darius. The whole group is all 6 olives, so 3/6 are green.**	The whole group is all 6 olives, so the denominator is 6. 3 of 6 are green: 3/6.

BOSS BATTLE: CLOSING THE KITCHEN (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Two steps! A big tray holds 12 olives in 2 equal rows. Owen fills one whole row with green olives. How many olives are green?	**6** (also accepted: six, 6 olives, 6 olive, 6 green olives)	Step 1: 12 olives in 2 equal rows is 12 ÷ 2 = 6 olives in each row. Step 2: one row is green, so 6 olives are green. That is 1/2 of the tray.
bb4_s0	Set 1: Find the mistake. A square pizza is cut into 6 equal slices. Hana takes 5 slices and says she took 5/30 of the pizza. What went wrong? [The pizza still has 6 equal parts, so 5 slices are 5/6. The denominator stays 6. / Nothing is wrong. 5 slices are 5/30. / 5 slices are 1/5 of the pizza. / 5 slices are 6/5 of the pizza.]	**The pizza still has 6 equal parts, so 5 slices are 5/6. The denominator stays 6.**	Each slice is 1/6. Counting 5 of them gives 5/6. The denominator stays 6, because the pizza is still cut into 6 equal parts.
bb3_s1	Set 2: Two steps! Darius has 15 garlic knots in 3 boxes, with the same number in each box. He sells one box. How many garlic knots did he sell?	**5** (also accepted: five, 5 garlic knots, 5 knots, 5 knot, 5 garlic knot)	Step 1: 15 ÷ 3 = 5 garlic knots in each box. Step 2: he sold one box, which is 1/3 of the knots, so he sold 5 garlic knots.
bb4_s1	Set 2: Find the mistake. Owen broke a breadstick into 3 pieces: one big piece and two tiny pieces. He ate the big piece and says he ate 1/3 of the breadstick. What went wrong? [The 3 pieces are not equal, so the big piece is not 1/3 of the breadstick. / Nothing is wrong. He ate 1 of 3 pieces, so he ate 1/3. / He ate 3/1 of the breadstick. / He ate 1/2 of the breadstick, because he ate 1 piece.]	**The 3 pieces are not equal, so the big piece is not 1/3 of the breadstick.**	Thirds must be 3 equal parts. A big piece and two tiny pieces are not equal, so the big piece is not 1/3.
bb3_s2	Set 3: Two steps! Hana sorts 16 napkins into 4 equal stacks. One whole stack is red. How many napkins are red?	**4** (also accepted: four, 4 napkins, 4 napkin, 4 red napkins)	Step 1: 16 ÷ 4 = 4 napkins in each stack. Step 2: one stack is red, so 4 napkins are red. That is 1/4 of the napkins.
bb4_s2	Set 3: Find the mistake. Owen says 3/4 of a square pizza means 3 + 4 = 7 parts. What went wrong? [3/4 means the whole has 4 equal parts, and we count 3 of them. We do not add. / Nothing is wrong. 3/4 means 7 parts. / 3/4 means 3 wholes and 4 more parts. / 3/4 means 4 − 3 = 1 part.]	**3/4 means the whole has 4 equal parts, and we count 3 of them. We do not add.**	3/4 means 3 of 4 equal parts. The whole has 4 parts, and we count 3 of them. We do not add the numbers.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W16.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 32 garlic knots in 4 equal boxes	32 ÷ 4 = 8 knots in each box.
2	Flatbread cut into 3 equal pieces	thirds
3	Draw a square pizza in 4 equal parts; shade 1	Drawing shows 4 equal parts with 1 shaded (1/4). Check that the parts are the same size.
4	2 big pieces and 2 small pieces: one fourth each?	No. The pieces are not equal, so they are not fourths.
5	6 olives, 2 green	6 olives in all, 2 are green, so 2/6 of the olives are green. Accept any reasonable attempt before the lesson.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Tray pizza in 6 equal parts; shade 1; unit fraction	Drawing shows 6 equal parts with 1 shaded; the unit fraction is 1/6.
2	8 peppers: 3 green, 2 yellow, rest red	Step 1: 3 + 2 = 5, and 8 − 5 = 3 red. Step 2: 3/8 are red.
3	Darius ate 2 of 6 slices and says 2/4	The denominator must be all 6 equal slices, not the 4 left. He ate 2/6 of the pizza.
4	What the 5 and the 8 tell in 5/8	The 8 (denominator) tells the pizza is cut into 8 equal parts; the 5 (numerator) tells 5 of those parts are counted.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Half of a 4-part pizza and half of an 8-part flatbread	2 parts of the pizza (2/4) and 4 parts of the flatbread (4/8) get basil.

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